



Case Study

Software Architecture
3rd Year – Semester 1
Assignment – 2

Schedule

- 1st March – Case Study Distribution
- 10th March – Topic Registration
 - Submission of Topic
 - Finalize Registration of Students for the Groups
- 11th March – 16th April – Progress Reviews
- 17th April – Presentation Upload (CourseWeb)
- 19th April (onwards) – Presentation & Viva

20 marks



What is a Case Study?

- An intensive analysis of an individual unit/item/situation
- Produce a detailed specialized report (non-generalized)
- You'll be conducting qualitative research to find specific details of how your subject is affected
- Study outcome may stress factors contributing to its success or failure

Eg: Civilian casualties in 9/11 attack



The Basics

- Subject: Choose the situation on which to write
- Facts: Gather as much information as possible about the situation
- Analyze: Analyze all of the elements surrounding the situation
- Evaluate: Determine the final solution implemented
- Conclude: Identify why the solution worked or did not work



Scope

- Narrow down the scope of your case

Eg:

- Education
- Education in Sri Lanka
- Higher Education in Sri Lanka
- Private Universities/Institutes in Sri Lanka
- ... more specific ...

Topic Selection

- This academic case study is about an intensive analysis about a Software System with respect to its **Architecture Style(s)** and **Key Quality Attribute(s)**

E.g. Microservice Architecture used by Facebook and its Scalability & Performance

- Software/System → Facebook
- Architecture Style(s) → Microservice
- Quality Attribute(s) → Scalability & Performance



Topic Selection – Rules

- 1 topic per group on First come first serve basis
- A group should propose only 1 topic at a time and should check if other groups have submitted the same/similar topic
- A group should have a Unique topic when the System, Architectural Style(s) and Key Quality Attribute(s) are combined for their study
- It's at the discretion of the Lecture to Approve/Reject the topic if...
 - Other groups have taken a similar topic
 - Case/Scope is not compelling enough to conduct a study

Topic Selection – Software System

- A Group must select a Software System from **one** of below categories:
 - VPN Systems
 - Storage Services
 - Artificial Intelligence / Machine Learning
 - Operating Systems
- Examples:

VPN Systems	Storage Services	AI/ML Systems	Operating Systems
• Nord VPN • Express VPN • Proton VPN	• Dropbox • Google Drive • OneDrive	• Chat GPT • Google AI • Open ML	• Linux • Windows • iOS

Expectations

- Case study should include background information on the specific topic, an analysis of the case under study showing problems or effective strategies, as well as recommendations
- Analyze
 - Why the system was built around the selected Architecture Style?
 - Why the identified the Quality Attributes are Key to the System success?
 - How the chosen Architecture Style and Quality Attribute(s) have improved or worsened overall Quality of the System under consideration?
- Your findings
 - What has been done to improve the chosen Quality Attribute(s) of the system
 - What other Quality Attributes were affected? and how so?
 - Did the System's Architecture evolve over time? If so why?
 - What are the Trade-offs made? What are the implications?
 - Your judgement/recommendations/suggestions
- Compare
 - Compare with competitor systems in the same domain and identify how the Architectural Style and Quality Attribute(s) under consideration helped to stay competitive (or why this particular system lost out to the others)

- Any type of plagiarism is not allowed.

Plagiarism: Academic honesty is crucial to a student's credibility and self-esteem, and ultimately reflects the values and morals of the Institute as whole. A student may work together with one or a group of students discussing assignment content, identifying relevant references, and debating issues relevant to the subject. Plagiarism occurs when the work of another person, or persons, is used and presented as one's own.

Registration

- 4-5 Students per group
 - Advised to keep the same group as Assignment #1
 - Exceptions are allowed on case by case basis
- Topic
 - Registration only for 1 topic
 - In case if the topic is rejected or taken by someone else the group is allowed to register again without an extension for the deadline
 - Later changes to the topic is highly discouraged and based on availability
- Submission of Topics
 - Only the Group Leaders should submit topics
 - Submission of topics only allowed via CourseWeb

Viva and Evaluation

- Although this is a group assignment, individuals will be assessed on their performance.
- Presentation Submission – via CourseWeb
- Presentation & Viva
 - Only the groups who have uploaded the Presentation on time will be allowed for the Presentation & Viva Session
 - Each member should be present at the Viva/Presentation
 - Lecture & Lab timeslots will be taken for the Viva/Presentation

Topic Registration

- Create a new Forum Discussion with the Topic
- Add details of Team Members
 - Add Student ID along with Name with Initials
 - Indicate Group Leader
- Provide Resource locations
 - Official Documentations
 - White papers
 - Other research/studies
- Lecturer would Approve/Reject the topic
- Upon Approval of the topic commence the assignment work

Assignment #2 - Topic Registration

Microservices Architecture used by Facebook and its Scalability & Performance

☒ Subscribed



Microservices Architecture used by Facebook and its Scalability & Performance

by Chathura.de Chathura Desilva - Saturday, 21 March 2020, 8:51 AM

Note: This topic was submitted as an example.

Team Members:

1. IT12341234 - A. B. C. Perera [Group Leader]
2. IT43214321 - X. Y. Wickramasinghe
3. IT56785678 - P. Q. R. Gamage
4. IT87658765 - G. H. Amarasinghe
5. IT98219821 - D. E. Herath

Resources for System/Architectural Documentation:

1. URL 1
2. URL 2
3. Attachments <Specify any Attachment document name>

[Permalink](#) | [Edit](#) | [Delete](#) | [Reply](#)



Re: Microservices Architecture used by Facebook and its Scalability & Performance

by Chathura.de Chathura Desilva - Saturday, 21 March 2020, 8:55 AM

This topic is approved.

Tags: **Approved**

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Recommended Reading

- <http://www.monash.edu.au/lls/llonline/quickrefs/27-case-study.pdf>
- <http://www.wikihow.com/Do-a-Case-Study>

Group Presentation Instructions

- Plan the presentation for 10 minutes
- Limit to a maximum of 20 slides
- Every team member must present their work
- Include relevant Student ID(s) on each slide
- Questions will be asked during the presentation and afterwards if required
- Bring a printout of the Presentation to the Viva session

Presentation Outline

Below is a “sample” distribution of Responsibilities among group members:

- Member 1:
 - Introduction (15 seconds), History and most significant changes to the Architecture, Current Architecture, Key strengths, weaknesses, areas to improve with reasons, Key findings & observations, other architectural styles in consideration, etc.
- Member 2:
 - Analysis on Quality Attribute #1, Reasons why its important for chosen system, what methods have been used to improve, relating the knowledge on patterns & tactics, etc., measurements/proof of how quality was improved, Trade-offs made when improving this quality, etc.
- Member 3:
 - Analysis on Quality Attribute #2 – can be the same as Member 2 but for a different quality attribute. Or split the scope of the analysis of 2-3 quality attributes between member 2 & 3 equally
- Member 4:
 - Identification of market needs (current & future), market competitors, what the system need to battle to gain market control (with respect to system qualities), benchmark how the competitors are performing with respect to selected quality attributes, suggestions, recommendations on improvement, alternatives, etc.

References

- Include further statistics & references at the end of the slide set. But plan not to present unless Interview panel asks questions on them



<Case Study Topic>

Date: <yyyy-mm-dd>

Group ID : <xxx-xxx-xxx>

System : <system name>

Architecture : <architecture style>

Key Qualities: <quality attributes>

Student ID	Name with initials

Heading

- Content

Charts & Graphs (if needed)

